

Conjecture 00000004391: the half graph is not the minimizer of the spectral radius

Statement and what is proved

Conjecture: among graphons of fixed edge density the maximum spectral radius is attained only by the twin-clique graphon and the minimum only by the half-graph graphon, both unique up to zero-measure modification. We refute the *minimum* clause, hence the conjunction. The maximum (twin-clique) clause is not formalized.

Definitions (as in the Lean statement block)

A graphon is a symmetric measurable $W : \mathbb{R}^2 \rightarrow [0, 1]$ (only $[0, 1]^2$ matters). Its density is $\int_0^1 \int_0^1 W(x, y) dy dx$. Its integral operator is $(T_W f)(x) = \int_0^1 W(x, y) f(y) dy$. A real λ is an eigenvalue if some $f \in L^2$ of $(0, 1]$, not a.e. zero, satisfies $T_W f = \lambda f$ a.e. on $(0, 1]$. Two readings of “spectral radius” are used: $\rho(W) = \sup\{\lambda\}$ (“largest eigenvalue”, Lean `spectralRadius`) and $\rho_{\text{abs}}(W) = \sup\{|\lambda|\}$ (`spectralRadiusAbs`); `sSup` of a set that is not bounded above would be 0 in Lean, so boundedness is proved (Lemma 1). Two half-graph conventions: $H_A(x, y) = \mathbf{1}[x + y \leq 1]$ (density $1/2$), and the bipartite half graph $H_B(x, y) = 1$ iff $(x \leq \frac{1}{2} \text{ and } x + \frac{1}{2} \leq y) \text{ or } (y \leq \frac{1}{2} \text{ and } y + \frac{1}{2} \leq x)$ (density $1/4$). The minimum clause for H and a radius ρ (Lean `MinPart`): for every graphon W with density equal to that of H , W minimizes ρ among such graphons iff $W = H$ a.e. on $[0, 1]^2$. Lean **Conjecture** is the disjunction of `MinPart` over the four (radius, convention) pairs; **Claim** = $\neg$ **Conjecture**.

Proof

Lemma 1 (bound). *If $W \leq c$ everywhere and λ is an eigenvalue, then $|\lambda| \leq c$. (Lean `eig_abs_le`.)*

Proof. Let $M = \int_0^1 |f| > 0$. Then $|T_W f(x)| \leq \int_0^1 W(x, y) |f(y)| dy \leq cM$, so $|\lambda| |f(x)| \leq cM$ a.e.; integrating, $|\lambda| M \leq cM$. $\square$

Lemma 2 (constant graphon). *For $p \in [0, 1]$ the constant graphon $W \equiv p$ has density p , eigenvalue p (take $f \equiv 1$), hence by Lemma 1 $\rho = \rho_{\text{abs}} = p$. (Lean `density_const`, `rad_const`, `radAbs_const`.)*

Lemma 3 (H_A). *H_A has density $1/2$ and, with $f(y) = \cos(\pi y/2)$, $Tf(x) = \int_0^{1-x} \cos(\pi y/2) dy = \frac{2}{\pi} \sin(\frac{\pi}{2}(1-x)) = \frac{2}{\pi} f(x)$ for $x \in (0, 1]$. Hence $\rho(H_A), \rho_{\text{abs}}(H_A) \geq 2/\pi > 1/2$. (Lean `density_half`, `half_eig`, `rad_half_ge`, `radAbs_half_ge`, `two_div_pi_gt`; $\pi < 4$ is `Real.pi_lt_four`.)*

Lemma 4 (H_B). H_B has density $\int_0^1 |x - \frac{1}{2}| dx = 1/4$. Let $g(y) = \cos(\pi y)$ for $y < \frac{1}{2}$ and $g(y) = \sin(\pi y - \frac{\pi}{2})$ for $y \geq \frac{1}{2}$. For $x \leq \frac{1}{2}$: $Tg(x) = \int_{x+1/2}^1 \sin(\pi y - \frac{\pi}{2}) dy = \frac{1}{\pi} \cos(\pi x) = \frac{1}{\pi} g(x)$. For $x > \frac{1}{2}$: $Tg(x) = \int_0^{x-1/2} \cos(\pi y) dy = \frac{1}{\pi} \sin(\pi x - \frac{\pi}{2}) = \frac{1}{\pi} g(x)$. Hence $\rho, \rho_{\text{abs}} \geq 1/\pi > 1/4$. (Lean `halfOpBip`, `density_bip`, `bip_eig`, `rad_bip_ge`, `radAbs_bip_ge`, `one_div_pi_gt`.)

Conclusion. Suppose `MinPart` ρH with $H \in \{H_A, H_B\}$ of density $p \in \{1/2, 1/4\}$. Applying the “iff” to $W = H$ (trivially $H = H$ a.e.) shows H minimizes ρ among graphons of density p ; in particular $\rho(H) \leq \rho(\text{const}_p) = p$, contradicting $\rho(H) > p$ from Lemmas 2–4. (Lean `not_minPart`, `main`.)

Lean theorem and audit

Main theorem: `C4391.main : C4391.Claim`, where `Claim := ¬ Conjecture`. File `lean/Conjecture4391/Basic.lean` (identical to the compiled `Tlmc/C4391.lean`). `#print axioms C4391.main`: `propext`, `Classical.choice`, `Quot.sound`. No `sorry`, no `native_decide`.

Scope of the refutation

Refuted: the minimum clause under both radius readings and both half-graph conventions above, with density fixed to that of the half graph. Not refuted: the maximum clause; half graphs of other densities or other normalizations; complex eigenvalues (the operator is self-adjoint with real spectrum, so real scalars are the standard reading). No literature is cited.